Youngjin Yang

Undergraduate Researcher | Life Sciences

yyjyyj1212@naver.com | GitHub | Personal Website

Research Interest

I am interested in RNA-mediated regulation of gene expression and in connecting cell-based experiments with computational analysis. My circRNA research has motivated me to investigate how sequence and RNA structure influence translation, and to develop research skills through hypothesis-driven experiments, quantitative comparisons, and critical interpretation of computational predictions.

Education

Chung-Ang University | 2026 - Present

B.S. in Life Sciences. GPA: 4.20 / 4.50.

Dongguk University | 2022.03 - 2024.02

Life Sciences. GPA: 4.09 / 4.50. Transferred to Chung-Ang University.

Combined GPA: 4.112 / 4.50.

Research Experience

Chung-Ang University - Kim Hak-gyun Lab | 2026

Research focus: circRNA translation optimization through IRES screening and UTR optimization.

Experimental work: HEK293T cell culture, transient transfection, molecular cloning, plasmid preparation, and dual-luciferase assays.

Analysis: Quantitative comparison of reporter measurements across candidates and repeated experiments; RNAfold-based exploration of SZ1 IRES variants.

Dongguk University - Im Gun-il Lab | Jun 2023 - Jan 2024

Cell-based research: Participated in regenerative medicine research, including FACS-based evaluation of cellular states in spheroid culture and bone-marrow preparation.

In vivo research: Participated in mouse and rabbit osteoarthritis-model experiments evaluating therapeutic cell- and drug-related effects, including GFP-based observation.

Selected Research Projects

circRNA Translation Optimization & Reproducibility Analysis | 2026

Participated in IRES screening in HEK293T cells using firefly and Gaussia luciferase reporter measurements. Compared candidate activity across repeated experiments and examined selected IRES-UTR combinations. Investigated SZ1 mutation-associated translation changes through structural comparisons and follow-up mutation and rescue experiments.

Selected Research Projects (continued)

Computational Analysis of SZ1 IRES Variants | 2026

Analyzed all 1,950 single-nucleotide substitutions across a 650-nt SZ1 IRES computationally using RNAfold-derived secondary-structure predictions. Examined changes in minimum free energy (ΔMFE), pairing state, and local structure relative to the reference sequence.

Linked structural features to translation-efficiency measurements from nine experimentally tested single-substitution variants to explore sequence-structure-function relationships, including exploratory machine-learning comparisons. The nine measured variants support hypothesis generation; this analysis does not establish a validated translation-efficiency predictor.

Used AI-assisted scripting and debugging to support data processing, feature analysis, and interactive visualization of the mutation landscape.

Technical Skills

Cell & Molecular Biology

HEK293T cell culture; transient transfection; Lipofectamine 3000; dual-luciferase assays; FACS; PCR; molecular cloning; plasmid preparation.

RNA & Biological Systems

circRNA engineering; IRES screening; UTR optimization; reporter assays; experimental design; reproducibility analysis.

Data, AI & Tools

Python with AI-assisted coding and debugging; quantitative and statistical analysis; RNAfold; structural feature analysis; data visualization; interactive mutation-landscape visualization; GitHub.

Leadership & Activities

Student Council | Undergraduate

Participated in student council activities for two years and served as a department/student council leader.

External & Campus Projects | Undergraduate

Participated in Korea Social Contribution Association supporter activities and led university projects including curriculum development and creative convergence projects.

Career Direction

I aim to deepen my understanding of gene-expression regulation by combining molecular and cell-based experiments with quantitative analysis. In an international research environment, I hope to learn new approaches to RNA biology, refine experimental hypotheses, and critically evaluate links between computational predictions and biological measurements.